COMPUTER ENGINEERING

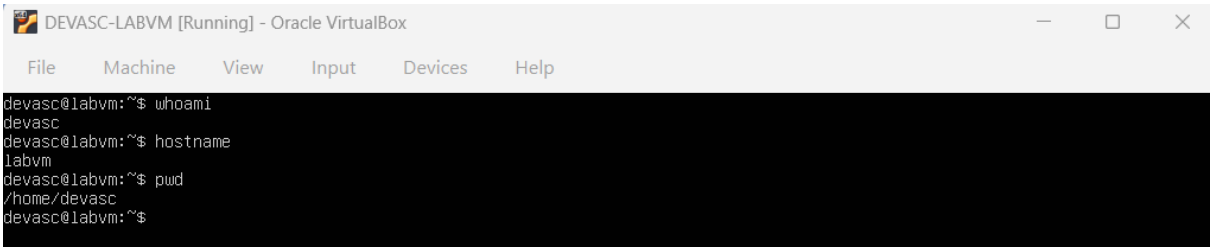
QUEZON CITY

COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S3
DATE PERFORMED:	10/29/2026	**DATE SUBMITTED:**	10/29/2026
NAME:	Pios, Joshua Paul B.	**INSTRUCTOR:**	Engr. Jimlord Quejado

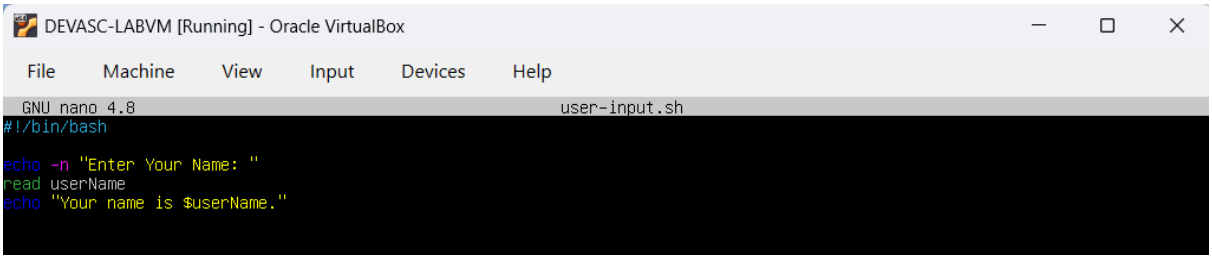
Seatwork 12.1 **Build a Sample Web App in a Docker Container**

1. PROCEDURE OUTPUT

DEVASC Environment



Bash script source code



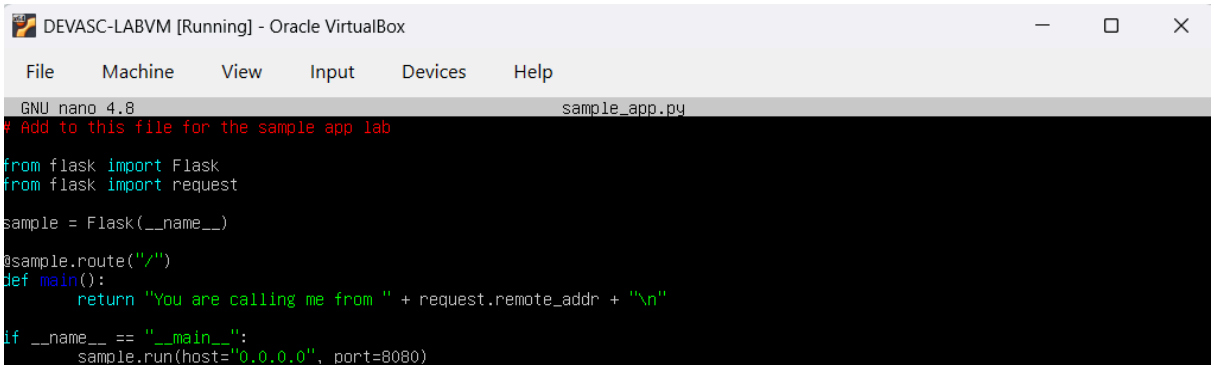
Successful Bash execution



Executable Bash script

```
devasc@labvm:~/labs/devnet-src/sample-app$ ls -l user-input.sh
-rw-rw-r-- 1 devasc devasc 86 Oct 27 19:09 user-input.sh
devasc@labvm:~/labs/devnet-src/sample-app$ chmod a+x user-input.sh
devasc@labvm:~/labs/devnet-src/sample-app$ ls -l user-input.sh
-rwxrwxr-x 1 devasc devasc 86 Oct 27 19:09 user-input.sh
devasc@labvm:~/labs/devnet-src/sample-app$ my user-input.sh user-input
my: command not found
devasc@labvm:~/labs/devnet-src/sample-app$ mv user-input.sh user-input
devasc@labvm:~/labs/devnet-src/sample-app$ ./user-input
Enter Your Name: Lajjie
Your name is Lajjie.
devasc@labvm:~/labs/devnet-src/sample-app$
```

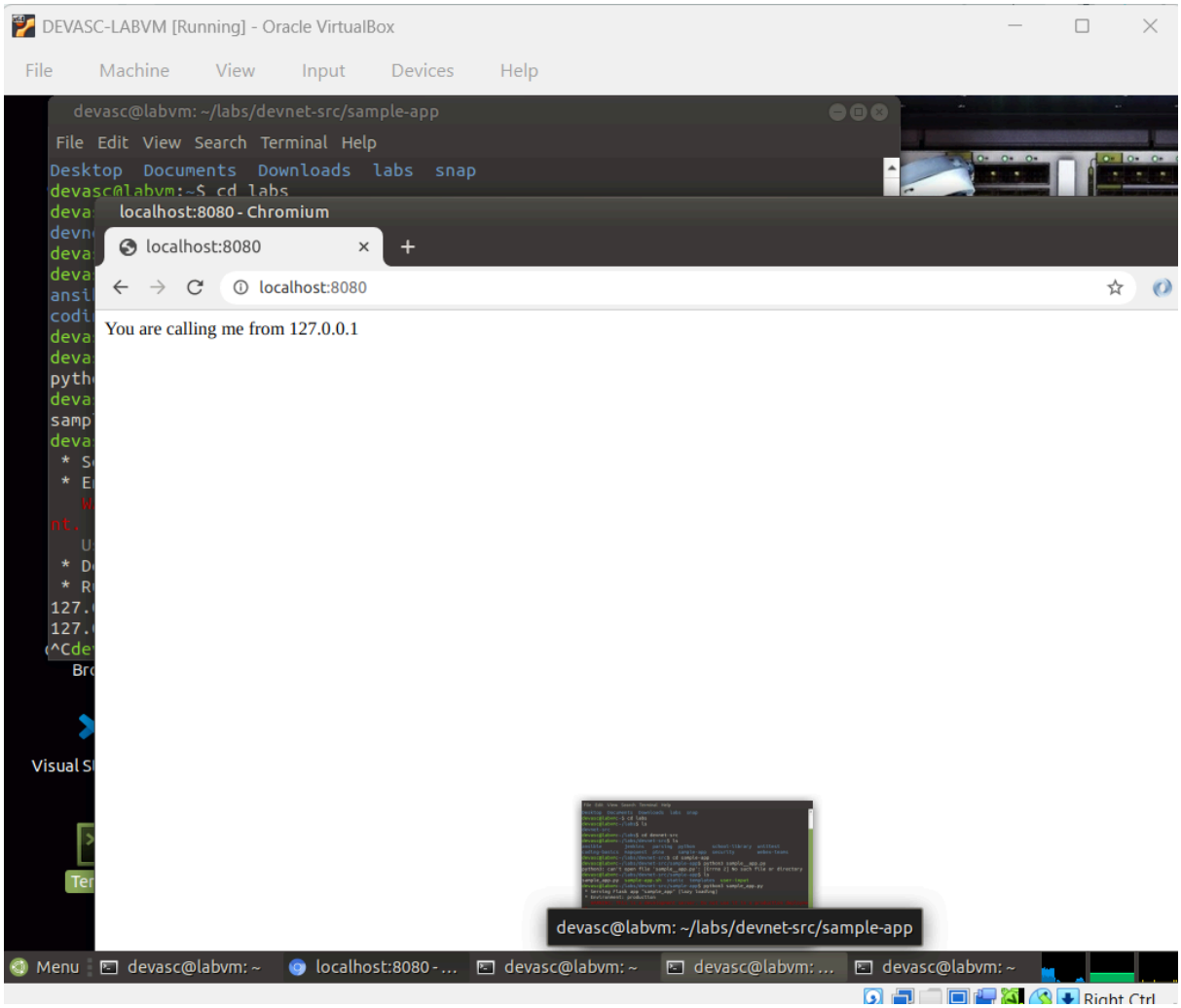
Initial Python Flask code



Flask server running

```
devasc@labvm:~/labs/devnet-src/sample-app$ python3 sample_app.py
 * Serving Flask app "sample_app" (lazy loading)
 * Environment: production
   WARNING: This is a development server. Do not use it in a production deployment.
   Use a production WSGI server instead.
 * Debug mode: off
 * Running on http://0.0.0.0:8080/ (Press CTRL+C to quit)
```

Initial website



HTML and CSS files

```
devasc@labvm:~/labs/devnet-src/sample-app$ cat templates/index.html
<html>
<head>
    <title>Sample app</title>
    <link rel="stylesheet" href="/static/style.css" />
</head>
<body>
    <h1>You are calling me from {{request.remote_addr}}</h1>
</body>
</html>
devasc@labvm:~/labs/devnet-src/sample-app$ cat static/style.css
body {background: lightsteelblue;}
devasc@labvm:~/labs/devnet-src/sample-app$
```

Updated Flask Python code

```
devasc@labvm:~/labs/devnet-src/sample-app$ cat sample_app.py
# Add to this file for the sample app lab

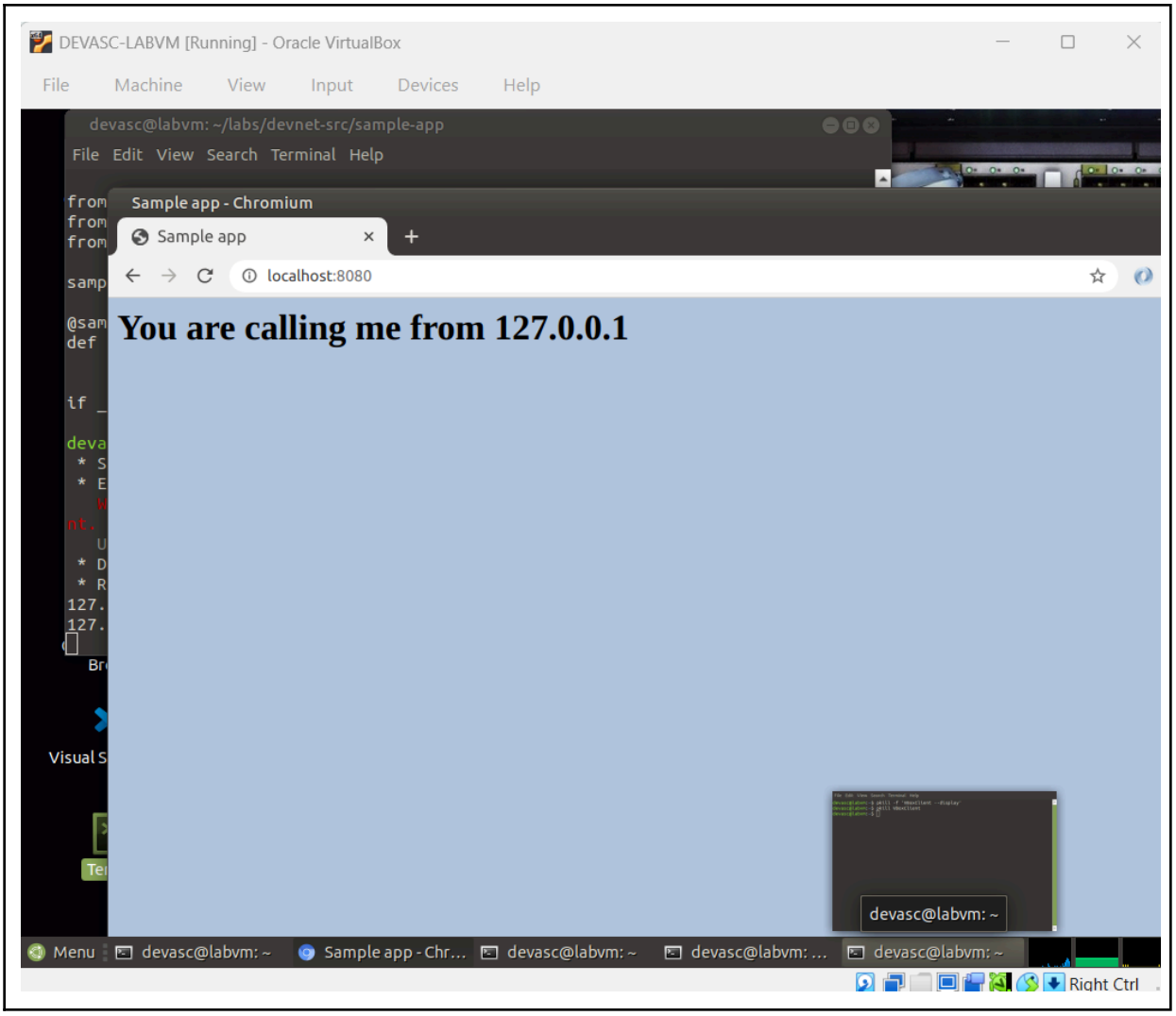
from flask import Flask
from flask import request
from flask import render_template

sample = Flask(__name__)

@sample.route("/")
def main():
        return render_template("index.html")

if __name__ == "__main__":
        sample.run(host="0.0.0.0", port=8080)
(devasc@labvm:~/labs/devnet-src/sample-app$
```

Styled web application



Docker automation Bash script

```
devasc@labvm:~/labs/devnet-src/sample-app$ cat sample-app.sh
#!/bin/bash
set -e

cd "$(dirname "$0")"

mkdir -p tempdir
mkdir -p tempdir/templates
mkdir -p tempdir/static

cp sample_app.py tempdir/.
cp -r templates/. tempdir/templates/
cp -r static/. tempdir/static/

cat > tempdir/Dockerfile <<'EOF'
FROM python:3.12-slim

WORKDIR /home/myapp

RUN pip install --no-cache-dir flask

COPY ./static /home/myapp/static/
COPY ./templates /home/myapp/templates/
COPY sample_app.py /home/myapp/

EXPOSE 8080

CMD ["python3", "/home/myapp/sample_app.py"]
EOF

cd tempdir

docker build -t sampleapp .

if docker container inspect samplerunning >/dev/null 2>&1; then
    echo "Container samplerunning already exists."
    exit 1
fi

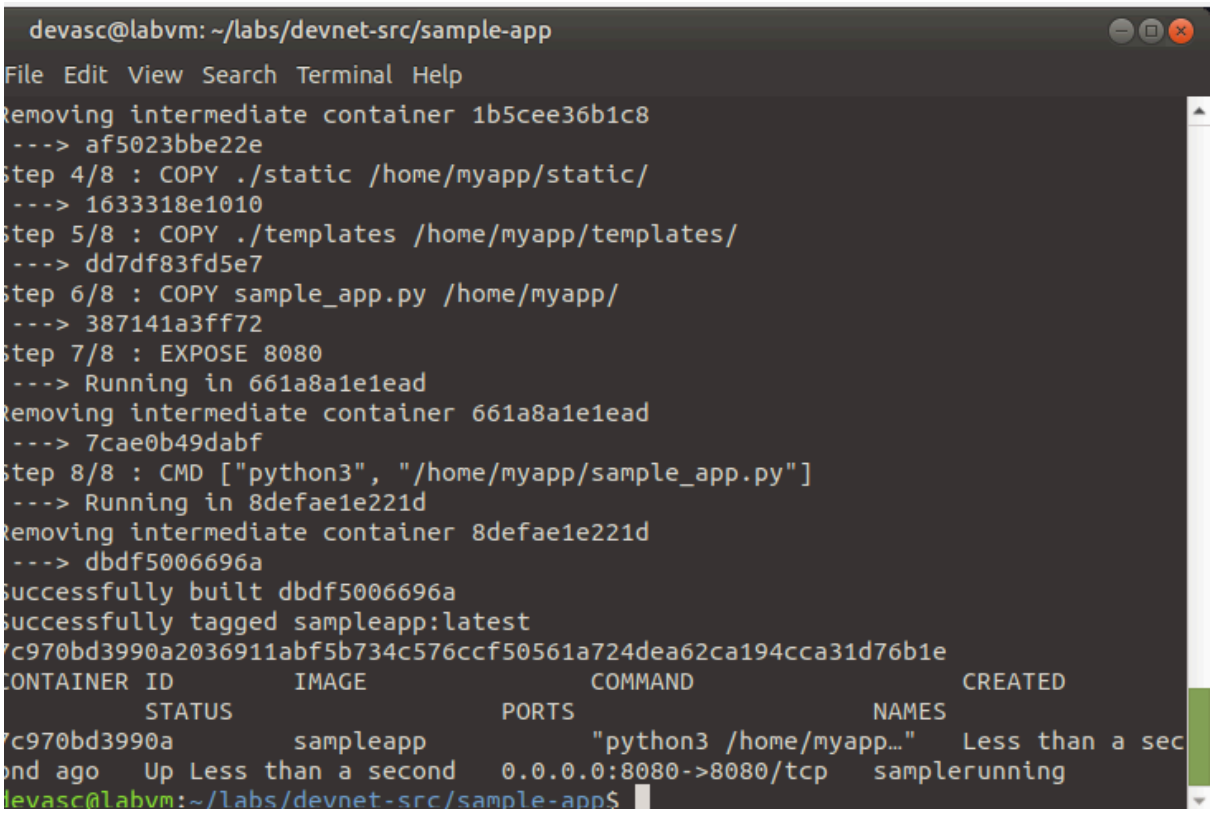
docker run -t -d -p 8080:8080 --name samplerunning sampleapp

docker ps -a
devasc@labvm:~/labs/devnet-src/sample-app$
```

Project file structure

```
devasc@labvm:~/labs/devnet-src/sample-app$ ls -l
total 20
-rw-r--r-- 1 devasc devasc  290 Oct 27 19:27 sample_app.py
-rwxr-xr-x 1 devasc devasc  741 Oct 27 19:05 sample-app.sh
drwxr-xr-x 2 devasc devasc 4096 Oct 27 19:25 static
drwxr-xr-x 2 devasc devasc 4096 Oct 27 19:25 templates
-rwxrwxr-x 1 devasc devasc   86 Oct 27 19:09 user-input
devasc@labvm:~/labs/devnet-src/sample-app$ ls templates/
index.html
devasc@labvm:~/labs/devnet-src/sample-app$ ls static/
style.css
devasc@labvm:~/labs/devnet-src/sample-app$
```

Successful Docker build



Generated Dockerfile

```
devasc@labvm:~/labs/devnet-src/sample-app$ ls tempdir/
Dockerfile  sample_app.py  static  templates
devasc@labvm:~/labs/devnet-src/sample-app$ cat tempdir/Dockerfile
FROM python:3.12-slim

WORKDIR /home/myapp

RUN pip install --no-cache-dir flask

COPY ./static /home/myapp/static/
COPY ./templates /home/myapp/templates/
COPY sample_app.py /home/myapp/

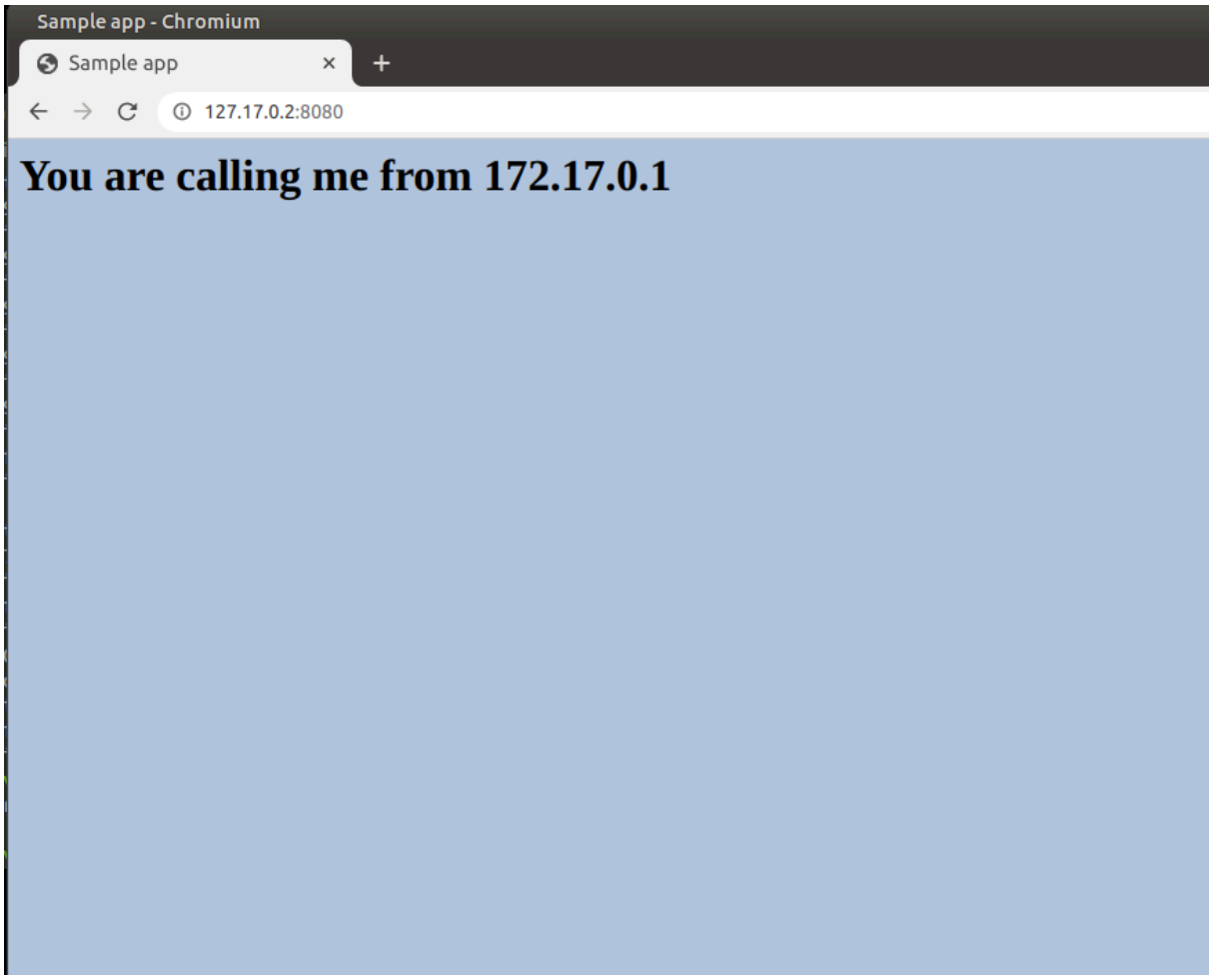
EXPOSE 8080

CMD ["python3", "/home/myapp/sample_app.py"]
devasc@labvm:~/labs/devnet-src/sample-app$
```

Running container information

```
devasc@labvm:~/labs/devnet-src/sample-app$ docker ps -a >> running.txt
devasc@labvm:~/labs/devnet-src/sample-app$ cat running.txt
CONTAINER ID        IMAGE               COMMAND                  CREATED
      STATUS              PORTS                    NAMES
7c970bd3990a        sampleapp           "python3 /home/myapp…"   About a minute
ago   Up About a minute   0.0.0.0:8080->8080/tcp   samplerunning
devasc@labvm:~/labs/devnet-src/sample-app$
```

Docker web application



Docker networking

```
ua       valid_lft forever preferred_lft forever
devasc@labvm:~/labs/devnet-src/sample-app$ docker inspect -f '{{range .NetworkSettings.Networks
}}{{.IPAddress}}{{end}}' samplerunning
172.17.0.2
devasc@labvm:~/labs/devnet-src/sample-app$
```

Inside the Docker container

```
devasc@labvm:~/labs/devnet-src/sample-app$ docker exec -it samplerunning /bin/bash
root@42e5c56e8992:/home/myapp# ls
sample_app.py  static  templates
root@42e5c56e8992:/home/myapp# ls /home/myapp
sample_app.py  static  templates
root@42e5c56e8992:/home/myapp# ls /home/myapp/templates
index.html
root@42e5c56e8992:/home/myapp# ls /home/myapp/templates/static
ls: cannot access '/home/myapp/templates/static': No such file or directory
root@42e5c56e8992:/home/myapp#
```

Stopped container

```
devasc@labvm:~/labs/devnet-src/sample-app$ docker stop samplerunning
samplerunning
devasc@labvm:~/labs/devnet-src/sample-app$ docker ps -a
CONTAINER ID        IMAGE               COMMAND                  CREATED             STATUS
                    PORTS               NAMES
42e5c56e8992        sampleapp           "python3 /home/myapp…"   4 minutes ago       Exited (13
7) 6 seconds ago                        samplerunning
devasc@labvm:~/labs/devnet-src/sample-app$
```

Docker container removed

```
devasc@labvm:~/labs/devnet-src/sample-app$ docker start samplerunning
samplerunning
devasc@labvm:~/labs/devnet-src/sample-app$ docker ps
CONTAINER ID        IMAGE               COMMAND                  CREATED             STATUS
          PORTS                    NAMES
42e5c56e8992        sampleapp           "python3 /home/myapp…"   4 minutes ago       Up 9 secon
ds        0.0.0.0:8080->8080/tcp   samplerunning
devasc@labvm:~/labs/devnet-src/sample-app$ docker stop samplerunning
samplerunning
devasc@labvm:~/labs/devnet-src/sample-app$ docker rm samplerunning
samplerunning
devasc@labvm:~/labs/devnet-src/sample-app$ docker ps -a
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS
     PORTS               NAMES
devasc@labvm:~/labs/devnet-src/sample-app$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: N/A

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: N/A

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: N/A

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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